

# **Data Analysis Report on Poverty Rate of India**

## **Introduction**

The objective of this data analysis and its report is to examine the distribution of people living in multi-dimensional poverty, with respect to the headcount of people and the change in the said headcount, along with the specific States and Union Territories and with the year the data was recorded.

## **DATA EXPLORATION**

### **How The Data Was Collected**

The dataset was collected from Open Government Data (OGD) Platform, India. It contains a summary of multi-dimensional poverty in India during two different periods, .i.e., 2013-2014 & 2023-2024.

### **Features Identified**

The features identified for the analysis is the headcount for the year 2013–2014 of people who were in multi-dimensional poverty, the headcount for the year 2023–2024 of people who were in multi-dimensional poverty, both gathered with respect to their States and Union territories, and also showing the total count of population that has escaped multi-dimensional poverty.

### **Data Visualization**

Python has been for Exploratory Data Analysis of the dataset. EDA was used to understand the dataset and its meanings and tellings more easily and

properly.

```
>> = RESTART: C:\Users\dhmi\OneDrive\Desktop\Pandas by me\poverty project\EDA.py =
count    2013_14_Headcount    2022_23_Headcount    escaped_multidimensional_poverty
mean      24.659200           7.751667           93.035600
std       14.761413           6.964059          139.620911
min        1.240000           0.370000           0.270000
25%       12.470000           2.762500           3.920000
50%       21.530000           5.425000          43.980000
75%       35.420000          11.175000          102.780000
max        56.340000          26.590000          593.690000
```

Fig 1: Statistical Outcome of all the values in the Dataset

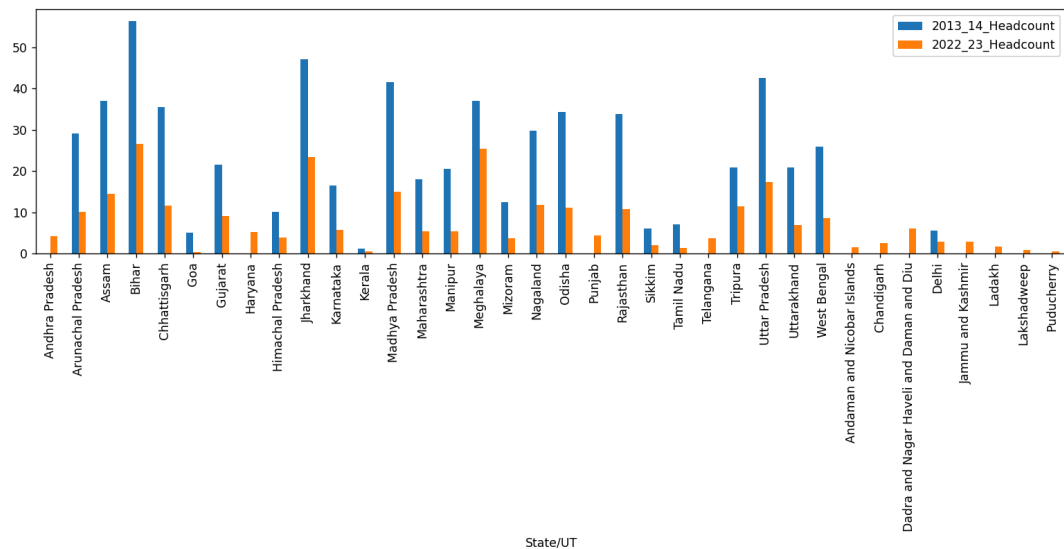


Fig 2: Showing the comparison of headcount per state/UT between different years. and UT.

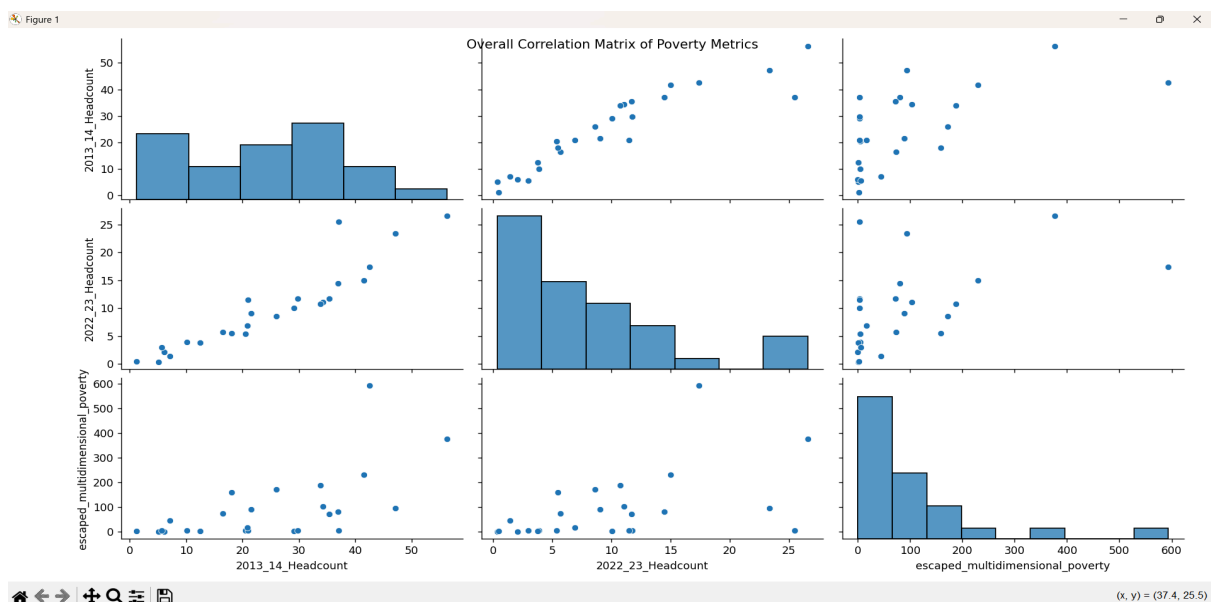


Fig 3: This is a pairplot (scatterplot matrix) comparing three poverty metrics 2013-14 headcount, 2022-23 headcount, and escaped multidimensional poverty across different states.

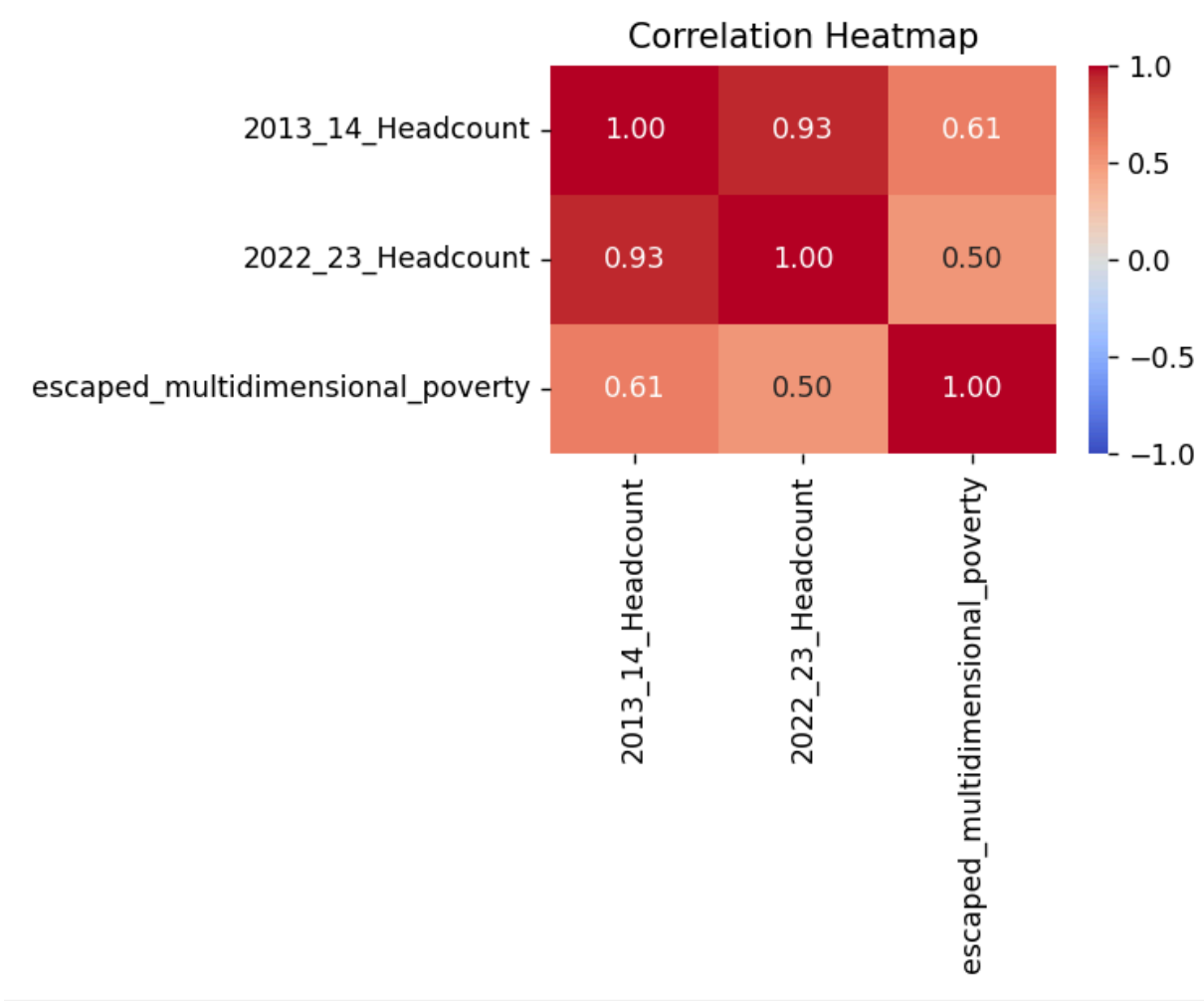
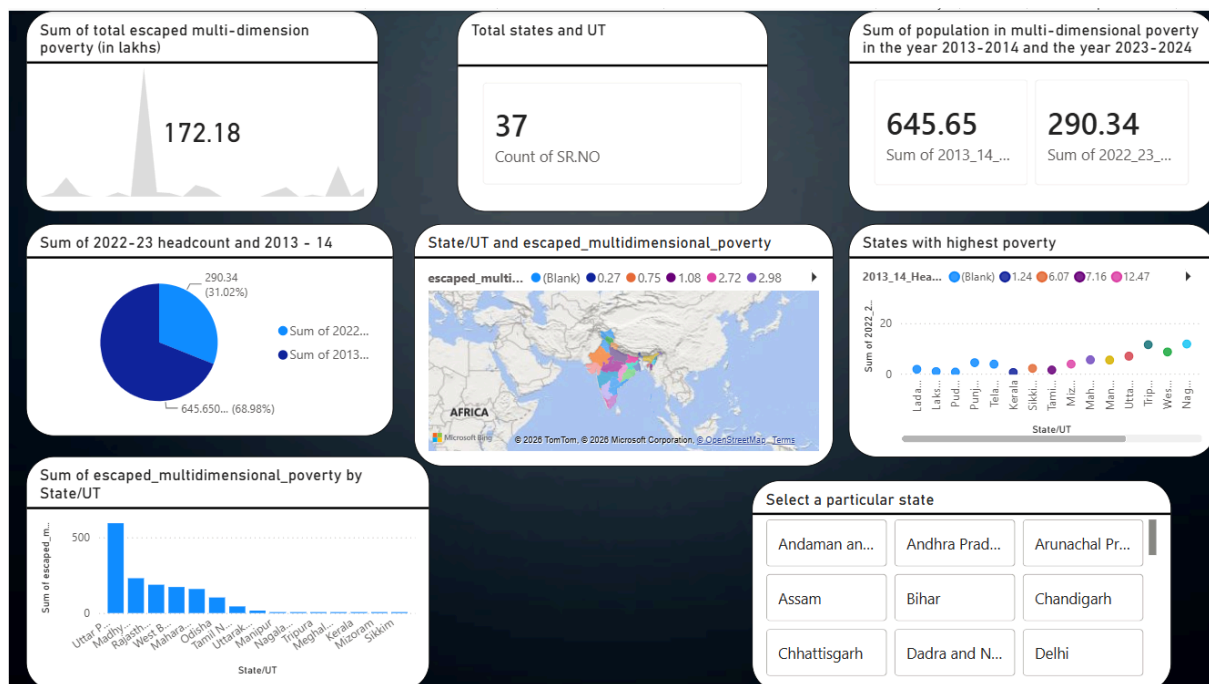


Fig 4 : 2013-14 and 2022-23 headcount poverty are very strongly correlated (0.93), confirming poverty levels track closely across the decade, while "escaped multidimensional poverty" correlates only moderately with both (0.61 and 0.50)

**Dashboard using Power BI for further analysis**



## Modeling and Results

The Multi-Dimensional Poverty dataset was used to create a dashboard and Exploratory Data Analysis (EDA) to understand the dataset better. This dashboard and EDA summarizes poverty trends across all states and Union Territories. It does not focus on one region. We have visualized values for each year and state through multiple charts. For the explanation part, we will start with the dashboard from bottom to the top

1. A state-level slicer was included. This allows users to drill down into one state or UT when needed.
2. The bar graph compares which state/UT has the highest count of people escaping poverty.
3. The scatter plot shows states/UT with the highest poverty rate.

4. The map visual plots each state and UT to show which state/UT and how many people escaped multi-dimensional poverty.
5. The pie chart shows which year has the highest headcount.
6. In Figure 1 Descriptive statistics show that the mean headcount ratio fell. It went from 24.66% in 2013–14 to 7.75% in 2022–23. About 93 lakh people per state and UT escaped multi-dimensional poverty.
7. In figure 2 and 3 Line and bar charts compare aggregate headcount ratios. The years compared are 2013–14 and 2022–23. This shows the total reduction in multi-dimensional poverty across the country over ten years.
8. In Figure 4 it shows a strong relationship ( $r = 0.93$ ) between the 2013–14 and 2022–23 headcount ratios, indicating that states with historically higher poverty levels have generally remained higher, even as poverty fell nationally. "Escaped multidimensional poverty" correlates only moderately with both headcount measures ( $r = 0.61$  and  $0.50$ ).

## Findings

Multi-Dimensional poverty declined substantially across India, between the years 2013–14 and 2022–23. The national average headcount ratio fell from 24.7% to 7.8%.

It has been observed that smaller states and UTs showed proportional reductions in poverty as compared to larger population dense states. It shows that there is a relation between geography and population density that affects the overall poverty of different states. Northern and Central states have a higher density of people living in multi dimensional poverty as compared to the south. It also means that progress has not been uniform among states to eradicate poverty.

## Conclusion

Although, there has been a decline in the total headcount between the years and that we do not have data for certain states, we can see that there has been a reduction of people living in multi-dimensional poverty. It is also noticed that the Southern part of India has a larger count of people escaping multi-dimensional poverty, compared to North or North-East. The GOI should ensure on creating newer policies for these places, so, that poverty reduction is similar all over India.

## References

1. Dataset by Open Government Data of India

Link:

<https://www.data.gov.in/resource/state-wise-details-multidimensional-poverty-india-estimates-2013-14-2022-23>